

Laxmi Devi Institute of Engineering and Technology, Chikani (Alwar)
 Department Name: Applied Science
 Program: B.Tech I-Year
 Branch & Sem. : All Branches & I
 Max. Time: 120 Mins

II-Mid Term (SET-1)
 Session: 2025-26
 Subject & Code: Mathematics I & 1FY1-01
 Max. Marks: 10

SECTION-A

Sec-A is compulsory and each part carry equal marks. (5 × 5 = 2.5)

- Q.1(i) Write the condition of exact for second order differential equation.
 (ii) If $1 - P + Q = 0$ write the part of C.F. known.
 (iii) State Eulers theorem.
 (iv) Write the second order differential equation.
 (v) If $P + Qx = 0$ write the part of C.F. known.

SECTION-B

Attempt any 2 question out of 3. (2 × 2 = 4)

Q.1. Solve the following differential equations:

$$x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = 2 \log x$$

Q.2. If $u = \sec^{-1} \left(\frac{x^3 + y^3}{x + y} \right)$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \cot u$

Q.3. Solve the following differential equation.

$$\frac{d^2 y}{dx^2} + (3 \sin x - \cot x) \frac{dy}{dx} + 2y \sin^2 x = e^{-\cos x} \sin^2 x$$

SECTION-C

Attempt any 1 question out of 2. (1 × 3.5 = 3.5)

Q.1. Solve the following differential equation by the method of Variation of parameter:

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^2 e^x$$

Q.2. If $u = f(r)$, where $r^2 = x^2 + y^2$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$

Laxmi Devi Institute of Engineering and Technology, Chikani (Alwar)
Department Name: Applied Science II-Mid Term
Program: B.Tech I-Year Session: 2025-26
Branch & Semester: All Branches & II Subject & Code: Engineering Chemistry
1FY2-03

Max. Time: 1:30 hours

Max. Marks: 10

SECTION-A

Sec-A is compulsory and each part carry equal marks. ($5 \times 5 = 2.5$)

Q.1 (i) Define Cement.

(ii) Define glass

(iii) Define corrosion

(iv) Glass manufacturing demands the following of how many steps? Write down the name of steps followed in manufacturing.

(v) Define Galvanisation

SECTION-B

Attempt any 2 questions out of 3. ($2 \times 2 = 4$)

Q.1. Discuss manufacturing of glass emphasizing the need of annealing

Q.2. Discuss important properties of lubricants.

Q.3. Discuss role of gypsum in manufacture of cement along with chemical reaction.

SECTION-C

Attempt any 1 question out of 2. ($1 \times 3.5 = 3.5$)

Q.1. Describe the wet process of manufacture of cement by rotatory kiln method with diagram.

Q.2. What is electrochemical corrosion? Describe the mechanism of electrochemical corrosion

Laxmi Devi Institute of Engineering and Technology, Chikani (Alwar)
Department of Applied Science
Midterm II Exam

Program: B.Tech Session: 2022-23 Subject & Code: CFP(1FY2-08)
Branch & Semester: All (Sem-1) Max Time: 1:30 Max Marks: 10
Faculty Name: Dr. Pratap Singh Patwal

PART A: Short answer questions (5 no. of questions 0.5 marks each).

1. Convert binary number 110101011000 to hexadecimal number
2. Explain sizeof operator with a suitable example.
3. Convert $(242)_{10}$ into hexadecimal.
4. Subtract 1101_2 and 1010_2 .
5. What is the difference between a compiler and an interpreter?

PART B: Analytical / Problem-Solving questions.

Attempt any two (2) questions out of four (04) ($2 \times 2 = 4$ marks)

1. Explain the concept of an Array in C.
2. Explain the function in C with a suitable example.
3. What is a string? Write a program to copy one string to another string without using the standard library function strcpy().
4. Write short notes on
 - a. Break
 - b. Continue
 - c. Goto
 - d. Conditional operator

PART C: Descriptive / Analytical / Problem-Solving / Design Questions.

Attempt one (01) question out of three (03) ($3.5 \times 1 = 3.5$ marks)

1. What is 2 D Array? Write a program to multiply two matrix's A and B of size 3×3 each and store the result in matrix C.
2. Write a program to pass matrix as an argument to a function and display the matrix within the function.
3. Convert the following numbers:
 - i. $(1056)_{16} = ()_8 = ()_2$
 - ii. $(5C6)_{16} = ()_8 = ()_{10}$
 - iii. $(74524)_8 = ()_8 = ()_{10}$
 - iv. $(10111)_2 = ()_{10} = ()_2$
 - v. $(4FEC)_{16} = ()_8 = ()_2$
 - vi. $(EFD)_{16} = ()_8 = ()_2$

Best of Luck.

Handwritten conversion table:

0	0	0	0	0
0	0	0	1	1
0	1	0		2
0	1	1		3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8

Laxmi Devi Institute of Engineering and Technology, Chikani (Alwar)

Department of Electrical Engineering

II-Midterm Test (Set-A)

Program:- B.Tech.

Subject & Code:-BEE (1FY2-07)

Session:- 2025-26

Max. Time:-90 min

Branch & Semester:-All(ISEM)

Max. Marks:-10

PART - A

➤ Attempt all questions. All questions carry equal marks.

(0.5x5=2.5)

Q.1 Define power factor.

Q.2 Define form factor for an ac wave.

Q.3 Define time period for an alternating quantity.

Q.4 What is depletion layer in semiconductor?

Q.5 Explain doping in semiconductor?

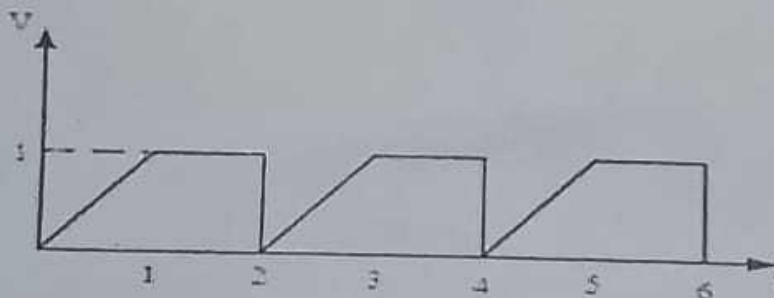
$$V = IR$$
$$I = \frac{V}{R}$$

PART - B

➤ Attempt any two questions. All questions carry equal marks.

(2x2=4)

Q.1 A periodic voltage waveform has been shown in figure, find out the average value and rms value.



Q.2 A series R-L circuit has an applied voltage $v = 150 \sin 500t$ with $R = 5$ and $L = 2\text{mH}$. Find Z , I and expression for i .

Q.3 Explain P-N junction diode in forward bias?

PART - C

➤ Attempt any one carry equal marks.

(3.5x1=3.5)

Q.1 Explain the response of circuit element (pure R, L & C) to ac voltage source.

Q.2 Explain working principal of NPN transistor?

Laxmi Devi Institute of Engineering and Technology
Department of Applied Science

II - Mid Term Exam -2025

B.Tech. 1st Semester (1st Year), Sub Code- 1FY3-05

Sub- Managerial Economics and Financial Accounting

Max Marks: 10

Part-A

Attempt all questions: -

(0.5*5=2.5marks)

1. What do you mean by demand?
2. Define price elasticity of demand.
3. Define cost.
4. Define demand schedule.
5. What is demand forecasting?

Part-B

Attempt any 2 questions: -

(2*2=4 marks)

6. Why price and demand have inverse relationship?
7. Explain the exceptions to the law of demand.
8. Explain the methods of demand forecasting.

Part-C

Attempt any 1 question: -

(1*3.5 = 3.5 marks)

9. Calculate breakeven point in terms of sales value and in units.

Fixed factory overheads cost -60,000

Fixed selling overheads cost -12000

Variable manufacturing cost per unit-12

Variable selling cost per unit-3

Selling price per unit-24

10. Define price elasticity of demand. Explain the methods to measure price elasticity of demand.